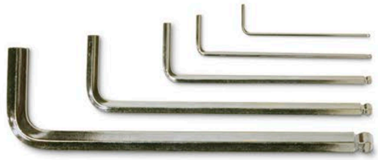
Allen wrenches in different sizes



Allen wrench

- **What?** Hexagonal socket wrench
- **Who?** William G. Allen
- **Where and when?** US, 1910

Created by the Allen Manufacturing Company in 1910, the Allen wrench is used to drive bolts and screws with hexagonal sockets in their heads. It is also known as a hex key. Allen wrenches drive screws right into the surface that they are fastening, keeping the surface smooth.

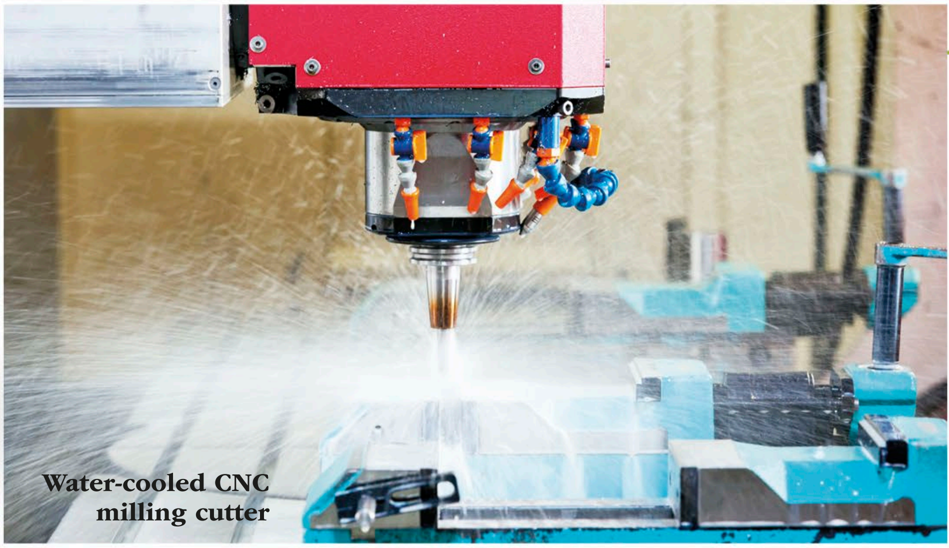
Phillips screwdriver

- **What?** Cross-headed screwdriver
- **Who?** Henry F. Phillips and Thomas M. Fitzpatrick
- **Where and when?** US, 1936

In the 1930s, Henry F. Phillips and Thomas M. Fitzpatrick invented cross-headed screws and screwdrivers. Cross-headed screws were particularly useful on automated car assembly lines, as they could take greater turning force and provided tighter fastening. With their cross-shaped tips, Phillips screwdrivers fit securely into screw heads.

The cross-shaped tip fits securely into the screw, making it easy to turn.

Modern Phillips screwdrivers



Water-cooled CNC milling cutter

CNC milling machine

- **What?** Computer-controlled cutting machine
- **Who?** John T. Parsons
- **Where and when?** US, 1940s

Milling is a process that uses a circular rotating cutter to cut into materials in several different directions, creating a variety of shapes. Milling machines have existed since the early 19th century, but in the 1940s, engineer John T. Parsons was the first person to consider using the earliest computers to control the milling process. CNC (computer numerical controlled) milling machines cut more precisely than manual machines.

Laser cutter

- **What?** Carbon dioxide laser
- **Who?** Kumar Patel
- **Where and when?** US, 1964

The laser, which produces a narrow, highly concentrated beam of light, was invented in the early 1960s. In 1964, engineer Kumar Patel discovered that carbon dioxide gas could create a laser beam that was intense and hot enough to cut through metal. Carbon dioxide lasers are still widely used today in cutting and welding, and for delicate surgical procedures such as eye surgery.

Metal laser-cutter

Laser level

- **What?** Laser level
- **Who?** Robert Genho
- **Where and when?** US, 1975

A laser level projects horizontal and vertical beams of light which can then be compared with a work surface. Laser levels are used in the construction industry so that builders can make sure they are working on perfectly horizontal surfaces or along straight lines.

Laser level on a building site